

BALA MURUGAN.R

Electrical and Electronics Engineering Student

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PROFESSIONAL SUMMARY

Motivated Electrical and Electronics Engineering student with a strong interest in electrical systems, embedded systems, control systems, EV technology, and automation. Interested in applying engineering knowledge to practical projects and developing reliable technical solutions. Seeking an entry-level opportunity to contribute technical skills while gaining industrial experience.

EDUCATION

Bachelor of Engineering – Electrical and Electronics Engineering 2023 – 2026

Ifet College Of Engineering, Villupuram/Tamil Nadu — CGPA: 7.89/10

Higher Secondary Education 2023

Alpha Matric Hr Sec School — Percentage: 61%

TECHNICAL SKILLS

- Electrical: Electrical Machines, Power Systems, Power Electronics, Control Systems, Circuit Analysis
- Embedded Systems: Sensors, Embedded Hardware, IoT Fundamentals
- Programming: Python, PLC, MATLAB
- Software/Tools: MATLAB/Simulink, AutoCAD, MS Office
- Other: EV Technology, Automation, Instrumentation, Renewable Energy

ACADEMIC PROJECTS

Satellite-Based Real-Time Forest Fire and Illegal Mining Detection System

Python | Image Processing | Satellite Technology (upcoming — I develop my own project)

- Developed a concept for monitoring forest regions using satellite imagery to identify potential forest fires and unauthorized mining activities.
- Designed an alert-based monitoring approach to notify relevant authorities when suspicious environmental changes are detected.
- Explored low-cost prototype implementation using image-processing and monitoring technologies.

Ultrasonic Fuel Tank Level Measurement System

Ultrasonic Sensor | Microcontroller | OLED

- Designed a system to measure fuel level using an ultrasonic distance sensor.
- Processed sensor measurements through a microcontroller and displayed the calculated level on an OLED display.
- Focused on providing a simple and low-cost solution for real-time fuel-level monitoring.

Automatic Temperature-Based Fan Controller

Temperature Sensor | Microcontroller | Motor Control

- Designed an automatic fan-control system based on surrounding temperature.
- Implemented sensor-based control logic to regulate fan operation according to temperature variation.

INTERNSHIP

Sundaram High Tech & Heavy Electricals 2025

Industrial Internship — Rampakkam, Tamil Nadu

- Gained practical exposure to industrial electrical systems, equipment, maintenance, and electrical engineering practices.
- Observed industrial processes and safety practices related to electrical and heavy electrical systems.

TVS Training Services Private Limited 2026

Industrial Internship

- Gained practical exposure to PLC programming, industrial automation, and control systems.
- Applied fundamentals of pneumatic systems, sensors, actuators, and electrical control components.

CERTIFICATIONS

- Getting Started with AI — IBM SkillBuild | 2026
- 3rd Prize in Technical Quiz — ISTE Chapter, Ifet College | 2026

- AI Course Completion — Infosys Springboard | 2026

ACHIEVEMENTS & ACTIVITIES

- Participated in technical events, workshops, and engineering-related learning activities.
- Developed academic projects focused on electrical, embedded, and emerging engineering technologies.

CORE STRENGTHS

Problem Solving | Technical Learning | Teamwork | Communication | Analytical Thinking | Project Development